PROBLEM STATEMENT NO. 9

AI Agent for Smart Farming Advice

Project Solution & Specification Document

1. THE CORE CHALLENGE

Empowering Grassroots Agriculture via Intelligence

Small-scale agricultural producers face massive vulnerabilities due to information gaps, unpredictable environments, and market volatility. This challenge details the engineering of an AI Agent designed to democratize critical data and deliver precise, real-time localized guidance straight to grassroots cultivators.

2. RETRIEVAL-AUGMENTED GENERATION (RAG) ARCHITECTURE

The planned solution utilizes a Retrieval-Augmented Generation infrastructure. Instead of relying solely on baseline static model knowledge, the agent dynamically references verified, trusted domain data sets during runtime execution.

Meteorological & Physical Inputs

Real-time localized weather updates, seasonal adjustments, and detailed local soil metrics.

Agritech Platform Streams

Dynamic crop recommendations, targeted pest control insights, and verified schedules from regional departments.

Live Mandi Market Tracking

Direct index lookups of standard commodity costs to prevent intermediary valuation exploitation.

Omnichannel Multi-Lingualism

Full support for local vernacular interfaces enabling conversational speech and text queries.

3. INTERACTION DESIGN & VALUE DELIVERABLES

Farmers interact intuitively using natural language inputs. The system processes variations in vernacular phonetics and returns structured answers targeted directly at optimization, risk mitigation, and crop yield enhancement.

Contextual Query Demonstrations:

• "What crop is best suited for this upcoming season based on current soil parameters?"

• "What is today's localized mandi market trade rate for tomatoes?"

By establishing data-driven checkpoints, the assistant reduces critical financial risk vectors, safeguards localized outputs, and elevates immediate household income across rural sectors.

MANDATORY TECHNOLOGY STACK STACK

- `IBM Cloud Lite Ecosystem Services`
- `IBM Granite Enterprise Foundation Models`